# Assignment 6 Report

## Part 1: Introduction

```
bash: duckdb: command not found...
aasha-devi@fedora:~/Documents/gmu-semester-8/assignment-6$ sudo curl https://install.duckdb.org | sh
[sudo] password for aasha-devi:
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  4163  100  4163    0     0  11781      0 --:--:-- --:--:-- --:--:-- 11826

*** DuckDB Linux/MacOS installation script, version 1.5.1 ***


         .;odxdl,
       .xXXXXXXXXXKc
       0XXXXXXXXXXXd  cooo:
      ,XXXXXXXXXXXXK  OXXXXd
       0XXXXXXXXXXXo  cooo:
       .xXXXXXXXXXKc
         .;odxdl,


######################################################################## 100.0%

Successfully installed DuckDB 1.5.1 to /home/aasha-devi/.duckdb/cli/1.5.1/duckdb
Updated symlink from /home/aasha-devi/.duckdb/cli/latest/duckdb to
                     /home/aasha-devi/.duckdb/cli/1.5.1/duckdb

Hint: Append the following line to your shell profile:
export PATH="/home/aasha-devi/.duckdb/cli/latest":$PATH
Also created a symlink from /home/aasha-devi/.local/bin/duckdb
                         to /home/aasha-devi/.duckdb/cli/latest/duckdb
```

## Part 2: Requirements

### Part 2a

1.
```
aasha-devi@fedora:~/Documents/gmu-semester-8/assignment-6$ duckdb
DuckDB v1.5.1 (Variegata)
Enter ".help" for usage hints.
memory D
```

2.

```
memory D .help
.bail on|off                      Stop after hitting an error.  Default OFF
.binary on|off                    Turn binary output on or off.  Default OFF
.cd DIRECTORY                     Change the working directory to DIRECTORY
.changes on|off                   Show number of rows changed by SQL
.columns                          Column-wise rendering of query results
.decimal_sep SEP                  Sets the decimal separator used when rendering numbers. Only for duckbox mode.
.databases                        List names and files of attached databases
.dump ?TABLE?                     Render database content as SQL
.display_colors [bold|underline]  Display all terminal colors and their names
.echo on|off                      Turn command echo on or off
.edit                             Opens an external text editor to edit a query.
.excel                            Display the output of next command in spreadsheet
.exit ?CODE?                      Exit this program with return-code CODE
.headers on|off                   Turn display of headers on or off
```

3.

```
memory D create table bank_transactions as (select * from read_csv('bank-transaction-data.csv', header = true, sep = ',', d
ateformat = '%m/%d/%Y'));
memory D .tables
 ───────── memory ─────────
 ────────── main ──────────

   bank_transactions

 Date              date
 Domain            varchar
 Location          varchar
 Value             bigint
 Transaction_count bigint

   1.00 million rows
```

4.

```
memory D describe bank_transactions;

   bank_transactions

 Date              date
 Domain            varchar
 Location          varchar
 Value             bigint
 Transaction_count bigint
```

5.

```
memory D explain analyze select count(*) "Total Records" from bank_transactions;
┌─────────────────────────────┐
│┌───────────────────────────┐│
││  Query Profiling Information  ││
│└───────────────────────────┘│
└─────────────────────────────┘
explain analyze select count(*) "Total Records" from bank_transactions;
┌─────────────────────────────────────┐
│┌───────────────────────────────────┐│
││         Total Time: 0.0011s         ││
│└───────────────────────────────────┘│
└─────────────────────────────────────┘
┌───────────────────────────┐
│           QUERY           │
└─────────────┬─────────────┘
┌─────────────┴─────────────┐
│      EXPLAIN_ANALYZE      │
│    ────────────────────   │
│                           │
│           0 rows          │
│           0.00s           │
└─────────────┬─────────────┘
┌─────────────┴─────────────┐
│    UNGROUPED_AGGREGATE    │
│    ────────────────────   │
│        Aggregates:        │
│        count_star()       │
│                           │
│                           │
│                           │
│           1 row           │
│           0.00s           │
└─────────────┬─────────────┘
┌─────────────┴─────────────┐
│         TABLE_SCAN        │
│    ────────────────────   │
│           Table:          │
│        memory.main        │
│     .bank_transactions    │
│                           │
│  Type: Sequential Scan    │
│                           │
│                           │
│                           │
│       1,004,480 rows      │
│           0.00s           │
└───────────────────────────┘
Run Time (s): real 0.001 user 0.001216 sys 0.001815
```

## Part 2b

1.

```
Run Time (s): real 0.008 user 0.028317 sys 0.011512
memory D select distinct domain from bank_transactions;
┌───────────────┐
│    Domain     │
│    varchar    │
├───────────────┤
│ MEDICAL       │
│ EDUCATION     │
│ INTERNATIONAL │
│ RESTAURANT    │
│ INVESTMENTS   │
│ PUBLIC        │
│ RETAIL        │
└───────────────┘
```

2.

a.

```
memory D select sum(value) from bank_transactions;
┌──────────────────┐
│   sum("value")   │
│      int128      │
├──────────────────┤
│   753207112952   │
│ (753.21 billion) │
└──────────────────┘
```

b.

```
memory D select sum(value) from bank_transactions where domain = 'RESTAURANT'
         ;
┌──────────────────┐
│   sum("value")   │
│      int128      │
├──────────────────┤
│   107498499345   │
│ (107.50 billion) │
└──────────────────┘
```

c.

```
memory D select sum(value) as "Total Value", domain from bank_transactions group by domain order by sum(value) desc;
┌──────────────┬───────────────┐
│ Total Value  │    Domain     │
│    int128    │    varchar    │
├──────────────┼───────────────┤
│ 107791432924 │ PUBLIC        │
│ 107790980756 │ MEDICAL       │
│ 107724396447 │ INTERNATIONAL │
│ 107658704394 │ EDUCATION     │
│ 107613592821 │ INVESTMENTS   │
│ 107498499345 │ RESTAURANT    │
│ 107129506265 │ RETAIL        │
└──────────────┴───────────────┘
```

d.

```
memory D select sum(value) as "Total Value", location from bank_transactions group by location order by location asc
        ;
┌─────────────┬───────────┐
│ Total Value │ Location  │
│   int128    │  varchar  │
├─────────────┼───────────┤
│ 16334511777 │ Ahmedabad │
│ 16450636199 │ Ajmer     │
│ 16337213682 │ Akola     │
│ 16484128413 │ Ambala    │
│ 16370012274 │ Amritsar  │
│ 16415501437 │ Ara       │
│ 16391505379 │ Banglore  │
│ 16375733132 │ Betul     │
│ 16402081340 │ Bhind     │
│ 16554394544 │ Bhopal    │
│ 16514534345 │ Bhuj      │
│ 16270208650 │ Bidar     │
│ 16211042348 │ Bikaner   │
│ 16507475172 │ Bokaro    │
│ 16404858470 │ Bombay    │
│ 16148860678 │ Buxar     │
│ 16423546290 │ Daman     │
│ 16329135604 │ Delhi     │
│ 16419577986 │ Doda      │
│ 16516102519 │ Durg      │
│      ·      │     ·     │
│      ·      │     ·     │
│      ·      │     ·     │
│ 16334597516 │ Kochin    │
```

i. There were more but I think this covers the execution of the correct sql command and its results

3.

a.

```
memory D select count(*) "Total Transactions Jan 4, 2022" from bank_transactions where date = make_date(2022, 1, 4)
        ;
┌────────────────────────────────┐
│ Total Transactions Jan 4, 2022 │
│             int64              │
├────────────────────────────────┤
│                           2752 │
└────────────────────────────────┘
```

b.

```
memory D select count(*) "Total Transactions Jan 4, 2022" from bank_transactions where date = make_date(2022, 1, 4) an
d domain = 'RETAIL';
┌────────────────────────────────┐
│ Total Transactions Jan 4, 2022 │
│             int64              │
├────────────────────────────────┤
│                            363 │
└────────────────────────────────┘
```

4.

a.

```
memory D select count(*) as "Total Transactions Jan 4, 2022", sum(value) as "Total Value" from bank_transactions where
 date = make_date(2022, 1, 4);
┌────────────────────────────────┬─────────────┐
│ Total Transactions Jan 4, 2022 │ Total Value │
│             int64              │   int128    │
├────────────────────────────────┼─────────────┤
│                           2752 │  2063604843 │
└────────────────────────────────┴─────────────┘
```

b.

```
memory D select count(*) as "Total Transactions March 2022", sum(value) as "Total Value" from bank_transactions where
date_part('year', date) = 2022 and monthname(date) = 'March';
┌───────────────────────────────┬─────────────┐
│ Total Transactions March 2022 │ Total Value │
│             int64             │   int128    │
├───────────────────────────────┼─────────────┤
│                         85312 │ 63876884769 │
└───────────────────────────────┴─────────────┘
```

```
memory D select count(*) as 'Total Transactions', sum(value) as 'Total Value', extract('year' from date) as Year, extract('month' from date) as Month from bank_transactions group by Year, Month order by Year desc, Month desc;
```

| Total Transactions<br>int64 | Total Value<br>int128 | Year<br>int64 | Month<br>int64 |
|---|---|---|---|
| 85312 | 63988218984 | 2022 | 12 |
| 82560 | 61927606668 | 2022 | 11 |
| 85312 | 63986742181 | 2022 | 10 |
| 82560 | 61911190101 | 2022 | 9 |
| 85312 | 64089898237 | 2022 | 8 |
| 85312 | 64049593378 | 2022 | 7 |
| 82560 | 61932318745 | 2022 | 6 |
| 85312 | 64015382961 | 2022 | 5 |
| 82560 | 61763054748 | 2022 | 4 |
| 85312 | 63876884769 | 2022 | 3 |
| 77056 | 57698986213 | 2022 | 2 |
| 85312 | 63967235967 | 2022 | 1 |

c.

5. Some information I, as a CEO, would like to obtain would be trends like maximum or minimum transactions within each month, as well as the months in which transactional values were highest.

```
memory D select min(value) as 'Smallest Transaction', max(value) as 'Biggest Transaction', extract('month' from date) as Month from bank_transactions group by Month order by Month desc;
```

| Smallest Transaction<br>int64 | Biggest Transaction<br>int64 | Month<br>int64 |
|---|---|---|
| 298434 | 1202265 | 12 |
| 298425 | 1202265 | 11 |
| 298428 | 1202269 | 10 |
| 298435 | 1202254 | 9 |
| 298435 | 1202268 | 8 |
| 298448 | 1202271 | 7 |
| 298425 | 1202264 | 6 |
| 298423 | 1202246 | 5 |
| 298440 | 1202239 | 4 |
| 298429 | 1202271 | 3 |
| 298432 | 1202233 | 2 |
| 298430 | 1202248 | 1 |

a.

```
memory D select avg(value) as 'Average Transactional Value',  extract('month' from date) as Month from bank_transactions group by Month order by avg(value);
```

| Average Transactional Value<br>double | Month<br>int64 |
|---|---|
| 748099.0158430233 | 4 |
| 748744.4294940922 | 3 |
| 748792.9066263497 | 2 |
| 749803.497362622 | 1 |
| 749893.2909520349 | 9 |
| 750032.1429693361 | 10 |
| 750049.4535821455 | 12 |
| 750092.1350290697 | 11 |
| 750149.2096051356 | 6 |
| 750367.8610394786 | 5 |
| 750768.8646145911 | 7 |
| 751241.3052911665 | 8 |

b.